SLU O607 DVT TEAM 17

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1. Introduction & Background

Project Context: The Opportunity dataset is an export from the platform's opportunity-management system. Each row represents a single opportunity (an internship, career track, competition, course, event, engagement, masterclass, job simulation, or program) that users can apply to. The raw file, Opportunity_Data_raw.xlsx, contained 5,733 rows and 33 columns covering identifiers, categorization, timing, cost, location, eligibility, and supporting content (descriptions, testimonials, tracking questions, and similar configuration fields stored as JSON text).

Problem Statement: The raw data (Opportunity_Data_raw.xlsx) is highly polluted with system noise, unescaped JSON export errors, inconsistent text conventions, and a massive footprint of QA/automation test records. It cannot be directly used for exploratory analysis or dashboarding without severe skewing of metrics. Before this dataset could be used for exploratory analysis or dashboarding, it needed a structured review: confirming what each column represents, identifying data quality problems, and applying consistent, documented fixes. That review and cleaning process is described below.

Project Objectives (Phase 1):

- Confirm semantic data definitions via a comprehensive column audit.
- Isolate, document, and resolve structural data flaws.
- Establish an analytical environment optimized for business intelligence without data loss.

2. Methodology & Data Preprocessing

Tools & Environment: MS Excel, Python, Google Docs, & Google Sheets

Data Quality Assessment & Profiling: The dataset was reviewed column by column, comparing value distributions, data types, and sample records against what each field was expected to represent. A complete Data Dictionary has been included as Sheet 2 of the cleaned workbook. It documents all 33 original variables, including their descriptions, data types, example values, and relevant notes, demonstrating a clear understanding of the dataset before the cleaning process began. Full detail on the cleaning logic follows

Data Retention Strategy

Preserved Fields (16 Columns)

Retained identity fields (opportunity_id, code), operational taxonomies (category, location), temporal markers, and financial metrics to calculate application windows and velocity. Columns were selectively preserved to ensure the final dataset remains optimized for analytical depth, modeling, and business intelligence without being bogged down by system noise.

Column	Definition
opportunity_id	Unique system identifier for each opportunity.
Category	Business category of the opportunity, such as Career or Internship.
code	Human-facing opportunity code.
created_at	Record creation timestamp stored as epoch milliseconds.
currency_type	Currency associated with fee or reward amounts.
Duration	Numeric duration value.
duration_type	Unit for duration, such as minutes, weeks, or years.
Fee	Participation or application fee amount.
is_auto_approve	Flag indicating whether approval is automatic.
last_date_to_apply	Application deadline timestamp stored as epoch milliseconds.
Location	Location or delivery mode, such as Virtual, WFM, or other text.
long_description	Detailed descriptive text about the opportunity.
Microscholarship	Scholarship or incentive amount tied to the opportunity.
modified_at	Last modified timestamp stored as epoch milliseconds.
Name	Opportunity title or name.
Short_description	Brief summary or teaser description of the opportunity.

Comprehensive Analysis of Retained Columns

Identity & Validation Fields (opportunity_id, code, name): opportunity_id serves as the primary system key required for exact de-duplication and row tracing. code acts as the human-facing identifier necessary for cross-referencing outside documents, while name provides the explicit business label essential for classification tasks.

Operational & Structural Taxonomy (category, location): category allows the business to segment entries into core tracks (such as Internships, Careers, or Competitions). location enables geographic or delivery-mode filtering (e.g., separating "Virtual" vs. "On-site" models).

Temporal Dimensions (created_at, modified_at, last_date_to_apply): These timestamp markers are critical for calculating velocity metrics, measuring application windows, and assessing record recency. Note: For these to be suitable for analysis, we had to convert them from raw epoch milliseconds into standardized date objects before visualization.

Financial & Control Systems (fee, microscholarship, currency_type, is_auto_approve): Combined, these fields outline the pricing barriers, incentive rewards, and transaction contexts of each opportunity. is_auto_approve serves as a vital workflow flag to measure operational automation versus manual moderation pipelines.

Excluded Fields from the Analytical Dataset (17 Columns)

Excluded redundant low-value metadata (**pk, summary**), highly sparse columns **(current_editor),** and broken relational JSON arrays that require external database schemas to decode (**Badge, Cohort, Reward**). These variables remain preserved in the original dataset but were excluded from the analytical dataset to improve analytical clarity and reduce unnecessary complexity. The final workspace generally falls into three problematic categories: low-value metadata, sparsity (high null counts), and broken relational linkages.

Comprehensive Analysis of Excluded Columns

1. Low Business Value & Redundant Data — The variables pk, image_link, and summary were omitted from the final dataset because they lacked analytical utility and introduced unnecessary redundancy.

2. High Null Values & Sparse Structural Metadata — The fields current_editor, is_archived, DropoutTransaction, and NotStartedTransaction were all excluded from the final dataset due to severe data sparsity and low analytical relevance. Specifically, **current_editor** and **is_archived** were dropped because their high volume of missing values and focus on temporary backend states or low-priority statuses offered little value to active reporting models. Similarly, **DropoutTransaction and NotStartedTransaction** suffered from significant data gaps, and because they merely represented sparse operational footprints with minimal standalone utility, they were removed to prevent introducing unnecessary noise into the rows.

3. Highly Critical But Broken Relational Entities — A massive hurdle encountered during data preparation was the handling of semi-structured objects that require external database schemas to decode. Despite their vital importance for building deep institutional insights, the following fields were dropped because they were exported as raw relational arrays or missing deep lookups. These include: **Badge, Cohort, Reward, CareerAddOn, Eligibility & Testimonial, Panellist, tracking_questions.**

3. Data Cleaning & Transformation

Removed corrupted rows. 3 rows had values shifted across the wrong columns (e.g. a date column containing text like "{ %22label%22: %22Ukraine%22"), consistent with an export error where an unescaped character inside a JSON field split a row into the wrong number of columns. These rows were not repairable with confidence, so they were removed from the working dataset and copied to a "Removed_Rows" reference sheet in the cleaned workbook.

Excluded incomplete records. Two records were excluded from the analytical dataset because they contained substantial missing information across several core analytical variables, including category, name, created_at, duration, and fee. Although these records contained valid system identifiers, the absence of sufficient business and operational information limited their analytical value and prevented meaningful interpretation. Consequently, these records were excluded from the analytical dataset to preserve analytical integrity, while the original dataset remained unchanged for reference and traceability.

Converted timestamps. *created_at, modified_at,* and *last_date_to_apply* were stored as Unix epoch milliseconds (e.g. 1710000000000). All three were converted to readable UTC date/time values.

Standardized currency_type. Values were trimmed and case-normalized, then mapped to one of USD, INR, or EUR (e.g. "usd", "USD" → USD; "EURO" → EUR). Values that didn't match a real currency (leftover JSON fragments from the export error) were set to blank rather than guessed at.

Standardized location. Common spelling and case variants were consolidated into single labels - for example "virtual" and "vitrual" both became "Virtual", and "work from home", "WFM", "wfh", and "WORK FROM HOME" all became "Work From Home". Other location values (e.g. named cities) were kept as entered, trimmed of extra whitespace.

Standardized duration_type. Singular and plural variants were consolidated ("week" and "weeks" → weeks; "month" and "months" → months; and similarly for days, hours, and minutes).

Standardized boolean fields. *is_archived* and *is_auto_approve* were converted to true True/False boolean values. Any entry that wasn't recognizably True or False (again, stray JSON-fragment artifacts) was set to blank.

Converted numeric fields. *fee, duration*, and *microscholarship* were converted from text to numeric types so they can be used directly in calculations (sums, averages, filters).

Standardized missing values. Text such as "null", "Null", "NaN", "n/a", "none", and empty strings appeared inconsistently across columns. All were normalized to a single, true blank cell so that missing-value counts and filters behave consistently across the whole dataset.

Standardized category values. Category labels were standardized to ensure consistent naming conventions (e.g., "JobSimulation" → "Job Simulation"), improving consistency for filtering, grouping, and analysis.

Appendix: Summary of the Final Dataset

Link to the cleaned Dataset:
https://docs.google.com/spreadsheets/d/1x4TpKp1jcSQAcB8mYFIqqkKIUZ9CxtEK/edit?usp=drivesdk&ouid=117688242522640077155&rtpof=true&sd=true

The cleaned dataset (Opportunity_Data_Cleaned.xlsx) contains 5,728 rows and 16 columns compared to the raw export; the cleaned version has:

- Consistent, human-readable dates in place of raw epoch timestamps.
- Consolidated, consistent category labels for currency, location, and duration units.
- True numeric and boolean types where applicable, instead of mixed text.
- A single, consistent representation of missing values across every column.
- No unrecoverable/corrupted rows.
- A transparent flag distinguishing likely test/automation records from real opportunity data, without discarding any rows.